NCR TVET COLLEGE NAMAKWALAND

91761 Occupational Certificate: Qualification: Electrician

SUMMATIVE ASSESSMENT

KM-01-KT01 – SAFETY, HEALTH, ENVIRONMENT, RISK AND QUALITY PRINCIPLES IN THE WORKPLACE

ASSESSMENT INSTRUMENT

SUMMATIVE: CANDIDATE'S ANSWERS			
NAME OF CANDIDATE		**NAME OF ASSESSOR**	
VENUE:		**TIME**	
TITLE	Occupational Certificate: Qualification: Electrician	**QUALIFICATION NUMBER**	91761
MARK POSSIBLE		**ACHIEVED:**	

EXAMINATION INSTRUCTIONS

1. The following Knowledge Assessment is based on the course in Occupational Certificate: Qualification: Electrician, Chapter KM-01-KT01: Safety, Health, Environment, Risk and Quality Principles in the Workplace
2. You may consult your Provider Participant's Guide or any other book or other documentation that is relevant to conducting assessment.
3. You may not have someone else do your examination for you. Your employer or supervisor must certify that the work is your own. Please remember to enclose a letter to this effect with your submission.
4. The examination counts out of 50 points. You must achieve at least 30 marks, or 60% to pass the knowledge questionnaire.
5. You have 2 hours to complete the assessment.
6. You are welcome to contact your facilitator or assessor by phone, fax or e-mail if you have any further questions about the examination. You may also make an appointment to visit the facilitator at the Provider premises.
7. The space left open for answers should be sufficient and gives you an indication of what is expected from you, at least in terms of how much you should write. Please do not enclose any annexures.
8. The marks allocated per question are given in brackets after each question.

Good Luck.

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Learning Institution: NCR TVET College Namakwaland

SECTION A: TRUE OR FALSE QUESTIONS: KNOWLEDGE APPLICATION

The primary purpose of exam true or false questions is to quickly determine a student's understanding of facts, address misconceptions, and generate information recall. These questions are designed to test students' knowledge of what has been taught in line with learning objectives. They promote active engagement with the material, requiring students to evaluate statements critically and identify key concepts. True or false questions are easy to grade, making them efficient for educators, especially in large classes. They can be graded rapidly, either manually or automatically, enabling teachers to quickly identify gaps in understanding.

ANSWER TRUE OF FALSE

Queston 1: KT0201 Legislation and regulations for workplace safety within electricity.	
Marks Possible	**10**

Q/no	Question	True/ False	Assessor	Moderator
1	Construction sites are considered one of the most dangerous places to work.	TRUE		
2	The Health and Safety at Work Act 1974 was created to promote safety in the construction industry.	TRUE		
3	Water extinguishers can be safely used on electrical fires.	False		
4	Employers are legally required to provide PPE free of charge to employees.	TRUE		
5	Employees must make full and proper use of PPE provided.	TRUE		
6	The right-hand grip rule explains how a magnetic field is established around a current-carrying conductor.	TRUE		
7	Overload protection devices safeguard motors against locked rotors, overloads, and short circuits.	TRUE		
8	Safety symbols and colour coding serve as reminders of the type of PPE required.	TRUE		
9	The Compensation for Occupational Injuries and Diseases Act (COIDA) applies to workplace injuries.	TRUE		
10	Keeping the workplace clean and tidy is a necessary safety precaution.	TRUE		
Marks Possible		10		
Mark Achieved				

ANSWER TRUE OF FALSE

Queston 2: KT0203 Safety precautions and safe practices for working within industry: KT0204 Personal protective equipment; KT0205 Safety symbols and colour coding	
Marks Possible	**28**

Q/no	Question	True/ False	Assessor	Moderator
1	Hazard identification is part of occupational health and safety?	TRUE		
2	Risk assessment is unnecessary in electrical installation work?	FALSE		
3	PPE includes protection for hands, face, ears, eyes, and feet?	TRUE		
4	Environmental protection concepts are part of occupational safety training?	TRUE		
5	Incident reporting is a key part of occupational health and safety?	TRUE		
6	Safety laws are unnecessary in modern workplaces?	FALSE		
7	Britain reports thousands of nonfatal injuries to the Health and Safety Executive each year?	TRUE		
8	It is a mistake to believe workplace injuries only happen in dangerous occupations?	TRUE		
9	The Compensation for Occupational Injuries and Diseases Act (COIDA) applies to workplace injuries?	TRUE		
10	A clean and tidy workplace is considered a safety precaution?	TRUE		
11	Waste materials must be disposed of properly to maintain safety?	TRUE		
12	Safety symbols and colour coding are used to indicate risks and required PPE?	TRUE		
13	Acts and regulations define the responsibilities of employers and employees?	TRUE		

Q/no	Question	True/ False	Assessor	Moderator
14	The inter-relationship between workplace safety and productivity is recognized in safety legislation?	TRUE		
15	Hazardous waste must be disposed of according to environmental regulations?	TRUE		
16	Alternators and generators are covered under protection devices training?	TRUE		
17	Alternators and generators are identical in construction and operation?	FALSE		
18	Overload protection devices are irrelevant to motor safety?	FALSE		
19	PPE is optional if an employee feels confident without it?	FALSE		
20	Fire extinguishers must be matched to the type of fire?	TRUE		
21	Electrical fires can be fought with water extinguishers?	FALSE		
22	The Health and Safety at Work Act requires employers to ensure safety at work?	TRUE		
23	Employees have no responsibility for their own safety?	FALSE		
24	Safety precautions include isolating and locking out equipment and circuits?	TRUE		
25	Securing worksites is part of fundamental safety practice?	TRUE		
26	Tools, equipment, and materials are included in safety training modules?	TRUE		
27	Occupational health and safety legislation applies to electricians?	TRUE		
28	PPE is designed to protect against risks to health and safety?	TRUE		

Q/no	Question	True/ False	Assessor	Moderator
Marks Possible		28		
Mark Achieved				

SECTION B: MULTIPLE CHOICE QUESTIONS

The purpose of multiple-choice questions (MCQs) in an exam is to efficiently measure a wide range of cognitive skills—from factual recall to complex data interpretation—while ensuring objective, standardized grading that eliminates examiner bias.

KEY PURPOSES AND ADVANTAGES INCLUDE:

1. Efficient Knowledge Assessment: They allow educators to test a broad breadth of course material in a single exam.
2. Objective Grading: They provide perfectly consistent, unbiased scoring that can be rapidly graded, often through automated systems.
3. Testing Higher-Order Thinking: When well-crafted, MCQs move beyond simple memory by requiring students to apply principles, analyse scenarios, and interpret data.
4. Pinpointing Misconceptions: By using plausible but incorrect answers (distractors), examiners can identify specific areas where students struggle or misunderstand concepts.
5. Rapid Feedback: Because they are easy to score, MCQs enable fast results and timely feedback for both instructors and learners

You will be presented by questions (referred to as the STEM) and four plausible answers (referred to as the Distractors), only one is correct - clearly mark your choice with an X.

Queston 3: Which type of fire extinguisher should NOT be used on electrical fires?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Dry powder extinguisher			
B	Foam extinguisher			
C	Water extinguisher (Correct)			
D	CO_2 extinguisher			

Queston 4: Why must employers provide PPE free of charge?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	To reduce company costs			
B	Because employees cannot buy their own			
C	It is a legal requirement under the Health and Safety at Work Act (Correct)			
D	To improve productivity only			

Queston 5: What is the purpose of incident reporting in the workplace?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	To punish employees for mistakes			
B	To record hazards and prevent future accidents (Correct)			
C	To reduce paperwork			

Option	Question	Your Answer	Assessor	Moderator
D	To avoid employer responsibility			

Queston 6: Which of the following is considered part of basic first aid training?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	Understanding safety symbols			
B	Performing CPR (Correct)			
C	Operating heavy machinery			
D	Writing incident reports			

Queston 7: Chapter 3: Measuring and Testing Instruments and Chapter 4: Fundamentals of Electricity: Portable instruments must be inspected, cleaned, and lubricated after use?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	A voltmeter is always connected in series with the circuit under test.			
B	Instrument transformers are used to step down current and voltage for safe measurement.			
C	Ohm's Law states that $V=I \cdot R$.			
D	Kirchoff's Law applies to both series and parallel circuits.			

Queston 8: Which instrument is used to measure electrical power in a circuit?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator

A	Ammeter			
B	Voltmeter			
C	Wattmeter (Correct)			
D	Frequency meter			

Queston 9: What is the correct connection method for a voltmeter?			
Mark Possible		**Mark Achieved**	

Option	**Question**	**Your Answer**	**Assessor**	**Moderator**
A	In parallel with the circuit (Correct)			
B	In series with the circuit			
C	Across the ammeter			
D	Across the transformer primary			

Queston 10: Which factor does NOT influence the resistance of a material?			
Mark Possible		**Mark Achieved**	

Option	**Question**	**Your Answer**	**Assessor**	**Moderator**
A	Length			
B	Cross-sectional area			
C	Temperature			
D	Colour (Correct)			

Queston 11: Which type of transformer is used for stepping down current for measurement?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	Step-up transformer			
B	Current transformer (Correct)			
C	Isolation transformer			
D	Auto transformer			

Queston 12: What is the primary purpose of incident reporting?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	To punish employees			
B	To reduce paperwork			
C	To prevent future accidents (Correct)			
D	To avoid employer responsibility			

Queston 13: Which instrument measures current?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	Voltmeter			
B	Ammeter (Correct)			
C	Wattmeter			

D	Frequency meter			

Queston 14: Which instrument measures voltage?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Ammeter			
B	Voltmeter (Correct)			
C	Wattmeter			
D	Power factor meter			

Queston 15: Which instrument measures electrical energy consumption?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Wattmeter			
B	Kilowatt-hour meter (Correct)			
C	Ammeter			
D	Frequency meter			

Queston 16: Which instrument measures cycles per second?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Wattmeter			
B	Frequency meter (Correct)			

C	Ammeter			
D	Voltmeter			

Queston 17: Which instrument measures the phase relationship between voltage and current?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	Wattmeter			
B	Power factor meter (Correct)			
C	Ammeter			
D	Voltmeter			

Queston 18: Which law defines the relationship between voltage, current, and resistance?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator
A	Kirchoff's Law			
B	Ohm's Law (Correct)			
C	Faraday's Law			
D	Newton's Law			

Queston 19: Which material is a conductor?			
Mark Possible	1	Mark Achieved	

Option	Question	Your Answer	Assessor	Moderator

A	Rubber			
B	Copper (Correct)			
C	Glass			
D	Plastic			

Queston 20: Which material is an insulator?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Copper			
B	Aluminium			
C	Rubber (Correct)			
D	Silver			

Queston 21: Which unit is used to measure resistance?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Volt			
B	Ampere			
C	Ohm (Correct)			
D	Watt			

Queston 22: Which unit is used to measure frequency?			
Mark Possible	**1**	**Mark Achieved**	

Option	Question	Your Answer	Assessor	Moderator
A	Hertz (Correct)			
B	Volt			
C	Ampere			
D	Ohm			

ASSESSMENT REVIEW

(to be completed by the Learners and Candidate Assessor)

LEARNER – PARTICULARS	
NAME OF LEARNER:	
QUALIFICATION NUMBER:	
TITLE OF QUALIFICATIOM:	
DATE OF REVIEW:	

REVIEW DIMENSION	CANDIDATE ASSESSOR	LEARNER 2	COMMENTS
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment instruments was fair, clear and understandable	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment judgements were made against set requirements.	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
Special needs were identified and assessment plan was adjusted.	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
Feedback was constructive against the evidence required.	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	

REVIEW DIMENSION	CANDIDATE ASSESSOR	LEARNER 2	COMMENTS
An opportunity to appeals was given.	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The evidence was recorded.	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	

DECLARATION BY LEARNER
POE 1/ LEARNER
I _____ (name and surname of candidate) am aware of the assessment and moderation process and understand that the moderator could declare the assessment decision invalid.

SIGNATURES			
POE 2/ LEARNER 2	**CANDIDATE ASSESSOR**	**PROVIDER ASSESSOR**	**INTERNAL MODERATOR**
DATE	**DATE**	**DATE**	**DATE**

SIGNATURE BY THE PROVIDER MODERATOR (THE SKILLS PROVIDER THAT TRAINED YOU – THE CANDIDATE ASSESSOR)			
NAME OF MODERATOR	**SIGNATURE BY MODERATOR**	**DATE OF MODERATION**	**REMARKS BY MODERATOR**

DECLARATION BY THE CANDIDATE
DECLARATION OF OWN WORK

I, Full Name: _____, RSA ID Nr:

_____, hereby confirm that the work submitted within this Workbook is my own.

_____ _____

SIGNATURE **DATE**

WITNESS TESTIMONY (To be completed by learner's Manager)

I, Full Name: _____, RSA ID Nr:

_____, being the (state appointment): _____

_____. of Company (state organization name),_____

_____, hereby confirm that the work submitted within this Workbook is that of the said learner and that the said learner consulted me for assistance in completing this Workbook on Questions relating to our business/ organisation.

_____ _____

SIGNATURE **DATE**